

Maximum Contiguous Pizza Sequence

Company: TCS

Round: Online

Year: 2022

Difficulty: Medium

Topic: Arrays, Sliding Window, HashMap

Source: <https://www.designa.in/8016/tcs-coding-questions-answers-october-2022-pizza-alice-2022>

Problem Description

Alice orders N pizzas. Each pizza has a flavor numbered from 1 to K .

The flavor of the i -th pizza is given by $A[i]$.

Bob can eat pizzas of at most $K - 1$ different flavors.

He wants to choose the maximum number of pizzas that are next to each other (contiguous) such that the selected pizzas contain at most $K - 1$ different flavors.

In Simple Words:

Find the longest continuous part of the array that contains at most $K - 1$ different flavors.

Example

Suppose:

$N = 7$

$K = 4$

$A = [1, 2, 3, 4, 2, 3, 2]$

Bob can eat at most:

$K - 1 = 3$

different flavors.

Consider:

$[2, 3, 4, 2, 3, 2]$

The different flavors are:

2, 3, 4

There are only 3 different flavors, so Bob can eat all 6 pizzas in this sequence.

Therefore, the answer is:

6

Input Format

From the visible part of the question:

- The input contains N , the number of pizzas.
- K represents the total number of possible flavors.
- The next N values represent the flavor of each pizza.
- $A[i]$ represents the flavor of the i -th pizza.

Output Format

Print a single integer representing the **maximum number of contiguous pizzas** that Bob can eat, such that the sequence contains **at most $K - 1$ different flavors**.

Sample Input

7 4
1 2 3 4 2 3 2

Sample Output

6

Explanation

$K - 1 = 3$, so Bob can eat pizzas having at most 3 different flavors.

The longest valid contiguous sequence is:

2 3 4 2 3 2

It contains only 3 flavors:

2, 3, 4.

Therefore, the maximum number of pizzas Bob can eat is:

6

Approach to Solve This Problem:

Use the Sliding Window technique.

1. Start with two pointers, left and right.
2. Move right through the array and add each pizza flavor to a frequency map.
3. Count how many different flavors are currently inside the window.
4. If the window contains more than $K - 1$ different flavors, move left forward and remove the flavors from the window.
5. Whenever the window has at most $K - 1$ different flavors, update the maximum length.
6. The maximum length obtained is the answer.

Java Solution:

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int k = sc.nextInt();
7         int[] a = new int[n];
8         for (int i = 0; i < n; i++) {
9             a[i] = sc.nextInt();
10        }
11        int limit = k - 1;
12        HashMap<Integer, Integer> freq = new HashMap<>();
13        int left = 0;
14        int maxLength = 0;
15        for (int right = 0; right < n; right++) {
16            freq.put(a[right], freq.getDefault(a[right], 0) + 1);
17            while (freq.size() > limit) {
18                freq.put(a[left], freq.get(a[left]) - 1);
19                if (freq.get(a[left]) == 0) {
20                    freq.remove(a[left]);
21                }
22                left++;
23            }
24            maxLength = Math.max(maxLength, right - left + 1);
25        }
26        System.out.println(maxLength);
27        sc.close();
28    }
29 }
```

Solution:

The main idea is to find the longest contiguous sequence of pizzas having at most $K - 1$ different flavors.

We use the Sliding Window technique:

1. Start with two pointers, left and right.
2. Move right forward and add the flavor of each pizza to a frequency map.
3. Count the number of different flavors in the current window.
4. If the window contains more than $K - 1$ flavors, move left forward until the window becomes valid again.
5. For every valid window, calculate its length:

$\text{right} - \text{left} + 1$

Complexity Analysis:

Time Complexity: $O(N)$

The right pointer moves from left to right once, and the left pointer also moves forward at most N times. Therefore, each element is processed a constant number of times.

Space Complexity: $O(K)$

The HashMap stores the frequency of the different pizza flavors present in the current window.